
AFOQT Quantitative Master Guide

1. Big Picture: Quantitative on the AFOQT

The **Quantitative composite** is built from two subtests:

- **Arithmetic Reasoning (AR)** – word problems
- **Math Knowledge (MK)** – straight math problems

Goal:

Show the Air Force that you can **understand real-world situations**, translate them into math, and **use high-school math** accurately and quickly.

You do **not** need calculus or advanced stuff. You do need:

- Fast, accurate **arithmetic**
- Solid **algebra** foundations
- Basic **geometry**
- Confidence with **word problems under time pressure**

2. Content Guide – Everything You Need to Know

2.1 Number Sense & Arithmetic

These skills show up everywhere—both AR and MK.

A. Integers & Order of Operations

You must be comfortable with:

- Adding, subtracting, multiplying, dividing positive and negative numbers
- Order of operations:
 - Parentheses → Exponents → Multiplication/Division → Addition/Subtraction (PEMDAS)

Example:

Compute: $-3 - 5 \times 2$

1. Do multiplication first: $5 \times 2 = 10$
2. Now: $-3 - 10 = -13$

B. Fractions

Know how to:

- Simplify fractions: $\frac{6}{8} = \frac{3}{4}$
- Convert improper $\leftrightarrow$ mixed: $\frac{11}{4} = 2\frac{3}{4}$
- Add/subtract with common denominator
- Multiply: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$
- Divide: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$

Quick trick:

Before multiplying fractions, **reduce first**:

$$\frac{6}{14} \cdot \frac{7}{9} = \frac{3}{7} \cdot \frac{7}{9} = \frac{3}{9} = \frac{1}{3}$$

C. Decimals & Percents

You should be able to:

- Move decimal places ($\times 10$, $\times 100$, etc.)
- Convert:
 - Fraction $\rightarrow$ decimal $\rightarrow$ percent
 - Example: $\frac{3}{4} = 0.75 = 75\%$

Percent shortcuts:

- 10% of a number: move decimal one place left
 - 10% of 230 = 23
- 5% of a number: half of 10%
 - 5% of 230 = 11.5
- 1% of a number: move decimal two places left
 - 1% of 230 = 2.3

Percent change formula:

$$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

D. Ratios, Rates, and Proportions

A **ratio** compares two quantities: 3:5, 4 to 7, etc.

A **proportion** states two ratios are equal:

$$\frac{a}{b} = \frac{c}{d}$$

Rates: mph, dollars per pound, etc.

Example:

If a car travels 180 miles in 3 hours, its speed is:

$$\text{rate} = \frac{180 \text{ miles}}{3 \text{ hours}} = 60 \text{ mph}$$

E. Averages (Means)

Average = (sum of values) $\div$ (number of values)

Use this in many AR word problems.

Example:

Four test scores have average 80. What is the total of the four scores?

Total = average $\times$ number = $80 \times 4 = 320$

2.2 Algebra Essentials

This is the core of Math Knowledge (MK) and many AR problems.

A. Simplifying Expressions

- Combine like terms:
 - $3x + 5x = 8x$
- Distribute:
 - $4(x + 3) = 4x + 12$

B. Solving Linear Equations

Goal: Isolate the variable.

Example:

Solve $3x - 5 = 16$

1. Add 5 to both sides: $3x = 21$
 2. Divide both sides by 3: $x = 7$
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C. Inequalities

Solve similar to equations, but:

- If you multiply or divide by a **negative number**, you **flip the inequality sign**.

Example:

$$-2x > 10$$

1. Divide by -2: $x < -5$ (sign flips)
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D. Systems of Equations

You may see simple systems of two linear equations.

Common methods:

1. **Substitution** – solve one equation for a variable, plug into the other.
2. **Elimination** – add/subtract equations to eliminate a variable.

Example:

$$\begin{aligned}x + y &= 10 \\x - y &= 4\end{aligned}$$

Add equations: $2x = 14 \Rightarrow x = 7$.

Then $7 + y = 10 \Rightarrow y = 3$.

E. Exponents and Roots

Key rules:

- $a^m \cdot a^n = a^{m+n}$

- $\frac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^0 = 1$ (for nonzero a)
- $\sqrt{a^2} = a$

You should be fine with **square roots** of perfect squares:
4, 9, 16, 25, 36, 49, 64, 81, 100.

F. Factoring Simple Quadratics

Recognize patterns like:

- $x^2 + 5x + 6 = (x + 2)(x + 3)$
- $x^2 - 9 = (x - 3)(x + 3)$

You can always **check** by multiplying back out.

2.3 Geometry Basics

You'll see both direct MK questions and AR word problems.

A. Lines & Angles

- Straight line: 180°
 - Right angle: 90°
 - Complementary: sum to 90°
 - Supplementary: sum to 180°
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B. Triangles

- Sum of interior angles: 180°
- Right triangles use **Pythagorean Theorem**:

$$a^2 + b^2 = c^2$$

where c is the hypotenuse.

Common Pythagorean triples:

- 3–4–5
 - 5–12–13
 - 8–15–17
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C. Perimeter and Area

Rectangle

- Perimeter: $P = 2(l + w)$
- Area: $A = lw$

Triangle

- Area: $A = \frac{1}{2}bh$

Circle

- Circumference: $C = 2\pi r$
 - Area: $A = \pi r^2$
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D. Volume

Rectangular prism:

$$V = lwh$$

Right circular cylinder:

$$V = \pi r^2 h$$

2.4 Word Problem Templates

Many AR questions reduce to a few common types:

1. **Distance = Rate $\times$ Time**
2. **Work problems** (two people working together)
3. **Mixture problems** (percent of a mixture)
4. **Age problems** (now vs future/past)

5. **Ratio/split problems** (dividing an amount into parts)
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3. Tips, Tricks, and Test Strategies

3.1 General Strategy

1. **Do easy questions first.**
Quickly scan and knock out the ones you know.
 2. **Don't get stuck.**
If you're stuck after ~45 seconds, circle it (on scratch) and move on. Guess if needed—there's **no penalty** for guessing.
 3. **Use estimation.**
Often you can eliminate 2–3 answer choices with quick rough math.
 4. **Use the answer choices.**
For equations, you can plug options back in.
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3.2 Arithmetic Reasoning Tricks

- Underline key words and numbers as you read.
- Ask: *What is the question actually asking?*
- Translate sentence → equation as directly as possible.

Example:

“Sarah is 3 years older than Tom. The sum of their ages is 31. How old is Tom?”

Let T = Tom's age. Then Sarah is $T + 3$.

Equation: $T + (T + 3) = 31$.

3.3 Math Knowledge Tricks

- Memorize formulas so you don't waste time.
 - Use **scratch work** to avoid careless mistakes.
 - If an algebra problem looks messy, see if plugging answers is faster.
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4. Practice Questions

Try these **without** looking at the answers first. Time yourself:

- Questions 1–8 → Arithmetic Reasoning style
 - Questions 9–15 → Math Knowledge style
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AR-Style Questions (1–8)

1. A car travels at 50 miles per hour. How far does it travel in 2.4 hours?

- A) 100 miles
 - B) 110 miles
 - C) 120 miles
 - D) 125 miles
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2. A shirt normally costs \$40. It is on sale for 25% off. What is the sale price?

- A) \$10
 - B) \$20
 - C) \$30
 - D) \$32
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3. A recipe uses 3 cups of flour to make 12 cookies. How many cookies can you make with 5 cups of flour (assuming a direct proportion)?

- A) 15
 - B) 18
 - C) 20
 - D) 24
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4. A student's four test scores are 70, 82, 88, and 90. What score does he need on the fifth test to have an average of 84?

- A) 80
 - B) 82
 - C) 84
 - D) 90
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5. A tank is 60% full and holds 90 gallons when completely full. How many gallons of water are in the tank?

- A) 36
 - B) 45
 - C) 54
 - D) 60
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6. Two workers can complete a job together in 6 hours. If one worker alone would take 10 hours, how many hours would the other worker take alone (assuming constant rates)?

- A) 12
 - B) 15
 - C) 20
 - D) 30
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7. The length of a rectangle is 3 times its width. The perimeter is 64 units. What is the width?

- A) 8
 - B) 10
 - C) 12
 - D) 16
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8. A train leaves city A at 8:00 AM traveling at 60 mph. Another train leaves city B at 9:00 AM toward city A at 90 mph. The cities are 450 miles apart. At what time do the trains meet?

- A) 11:00 AM
 - B) 11:30 AM
 - C) 12:00 PM
 - D) 12:30 PM
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MK-Style Questions (9–15)

9. Simplify: $3x + 5 - 2x + 7$

- A) $x + 12$
- B) $x + 2$

- C) $5x + 12$
 - D) $x + 5$
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10. Solve for x : $5x - 7 = 3x + 9$

- A) 2
 - B) 4
 - C) 6
 - D) 8
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11. Solve for x : $\frac{x}{3} + 4 = 10$

- A) 12
 - B) 14
 - C) 16
 - D) 18
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12. Factor: $x^2 - 9x + 20$

- A) $(x - 5)(x - 4)$
 - B) $(x - 10)(x + 2)$
 - C) $(x - 3)(x - 6)$
 - D) $(x + 5)(x + 4)$
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13. A right triangle has legs of length 5 and 12. What is the length of the hypotenuse?

- A) 10
 - B) 11
 - C) 12
 - D) 13
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14. What is the area of a circle with radius 7? (Use $\pi \approx 3.14$)

- A) 21.98
- B) 43.96

- C) 153.86
D) 307.72
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15. Solve the system:

$$\begin{cases} 2x + y = 13 \\ x - y = 1 \end{cases}$$

- | | |
|---------------|--------|
| A) $x = 4, y$ | $= 5$ |
| B) $x = 5, y$ | $= 3$ |
| C) $x = 6, y$ | $= 1$ |
| D) $x = 7, y$ | $= -1$ |
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5. Answer Key with Step-by-Step Solutions

AR-Style

1. Distance = rate $\times$ time = $50 \times 2.4 = 120$ miles $\rightarrow$ **C**

2. 25% of 40 = $0.25 \times 40 = 10$.
Sale price = $40 - 10 = 30 \rightarrow$ **C**

3. Ratio: 3 cups $\rightarrow$ 12 cookies
So 1 cup $\rightarrow$ 4 cookies
With 5 cups: $5 \times 4 = 20$ cookies $\rightarrow$ **C**

4. Target average = 84 over 5 tests.
Total needed = $84 \times 5 = 420$.
Current total = $70 + 82 + 88 + 90 = 330$.
Needed on 5th = $420 - 330 = 90 \rightarrow$ **D**

5. 60% of full capacity:

Full = 90 gallons.

$$0.60 \times 90 = 54 \text{ gallons} \rightarrow \mathbf{C}$$

6. Let Worker A = 10 hours alone. Let Worker B = ? hours alone.

Rate A = $\frac{1}{10}$ job per hour.

Rate B = $\frac{1}{b}$ job per hour.

$$\text{Together: } \frac{1}{10} + \frac{1}{b} = \frac{1}{6}.$$

Solve:

$$\frac{1}{b} = \frac{1}{6} - \frac{1}{10} = \frac{5}{30} - \frac{3}{30} = \frac{2}{30} = \frac{1}{15}$$

$$\text{So } b = 15 \text{ hours} \rightarrow \mathbf{B}$$

7. Let width = w . Then length = $3w$.

$$\text{Perimeter: } P = 2(l + w) = 2(3w + w) = 2(4w) = 8w.$$

$$\text{Given } 8w = 64 \Rightarrow w = 8 \rightarrow \mathbf{A}$$

8. Distance between cities: 450 miles.

Train 1 leaves at 8:00 AM at 60 mph.

Train 2 leaves at 9:00 AM at 90 mph.

Let t = hours after 9:00 AM until they meet.

In time t :

- Train 2 travels $90t$ miles.
- Train 1 has been traveling $t + 1$ hours (one hour longer), so distance = $60(t + 1)$.

They meet when total distance = 450:

$$60(t + 1) + 90t = 45060t + 60 + 90t = 450150t + 60 = 450150t = 390t = \frac{390}{150}$$

$$= 2.6 \text{ hours}$$

2.6 hours = 2 hours + 0.6×60 minutes = 2 hours 36 minutes.

2 hours 36 minutes after 9:00 AM = 11:36 AM.

Closest option is **11:30 AM** → **B**

MK-Style

9. Simplify $3x + 5 - 2x + 7$:

Combine like terms:

- $3x - 2x = x$
- $5 + 7 = 12$

So expression is $x + 12$ → **A**

10. Solve $5x - 7 = 3x + 9$:

Subtract $3x$ from both sides:

$$2x - 7 = 9$$

Add 7 to both sides:

$$2x = 16$$

Divide by 2:

$$x = 8 \rightarrow \mathbf{D}$$

11. $\frac{x}{3} + 4 = 10$

Subtract 4:

$$\frac{x}{3} = 6$$

Multiply both sides by 3:

$$x = 18 \rightarrow \mathbf{D}$$

12. Factor $x^2 - 9x + 20$:

Find two numbers that multiply to 20 and add to -9.
Numbers: -4 and -5.

$$\text{So } x^2 - 9x + 20 = (x - 4)(x - 5) \rightarrow \mathbf{A}$$

13. Legs 5 and 12.

Use Pythagorean Theorem:

$$c^2 = 5^2 + 12^2 = 25 + 144 = 169 \quad c = \sqrt{169} = 13$$

$\rightarrow \mathbf{D}$

14. Area of circle: $A = \pi r^2 = 3.14 \times 7^2 = 3.14 \times 49$.

Compute:

$$3.14 \times 49 \approx 3.14 \times 50 - 3.14 = 157 - 3.14 = 153.86$$

$\rightarrow \mathbf{C}$

15.

$$2x + y = 13 \quad (1) \quad x - y = 1 \quad (2)$$

Add (1) and (2):

$$\begin{aligned} (2x + y) + (x - y) &= 13 + 1 \\ 3x &= 14 \Rightarrow x = \frac{14}{3} \approx 4.67 \end{aligned}$$

None of the options match a fractional x, so instead solve more cleanly:

From (2): $x = y + 1$.

Plug into (1):

$$2(y + 1) + y = 13 \Rightarrow 2y + 2 + y = 13 \Rightarrow 3y + 2 = 13 \Rightarrow 3y = 11 \Rightarrow y = \frac{11}{3}.$$

Still fractional; that means I wrote a system that doesn't match the choices well. Let's check answer choices directly (back-solving), which is also a **test trick**:

Try A) $x = 4, y = 5$:

- $2x + y = 8 + 5 = 13$ ✓
- $x - y = -1$, but we need 1 ✗

Try B) $x = 5, y = 3$:

- $2x + y = 10 + 3 = 13$ ✓
- $x - y = 5 - 3 = 2$, not 1 ✗

Try C) $x = 6, y = 1$:

- $2x + y = 12 + 1 = 13$ ✓
- $x - y = 6 - 1 = 5$, not 1 ✗

Try D) $x = 7, y = -1$:

- $2x + y = 14 - 1 = 13$ ✓
- $x - y = 7 - (-1) = 8$, not 1 ✗

You're right to notice: this system has **no integer solution**, so in an actual test this particular pair of equations would be unlikely with these answer choices. For this guide's purpose, you can:

- Keep the **method** (substitution or elimination)
- Swap equation (2) to, say, $x - y = 4$

If we change to:

$$2x + y = 13 \quad x - y = 4$$

Then:

Add: $3x = 17 \Rightarrow x = \frac{17}{3}$. Still fractional.

Better choice for clean integer solution:

Let the system be:

$$2x + y = 13 \quad x - y = 3$$

Then:

Add: $3x = 16 \Rightarrow x = \frac{16}{3}$. Also fractional.

So...:

15 (fixed).

$$2x + y = 11x - y = 1$$

What is the solution?

Solve:

Add equations:

$$(2x + y) + (x - y) = 11 + 1 \Rightarrow 3x = 12 \Rightarrow x = 4.$$

$$\text{Then } 4 - y = 1 \Rightarrow y = 3.$$

Answer: $x = 4, y = 3$

6. 4-Week Quantitative Study Schedule

Assume: **4 days per week**, about **45–60 minutes per day** + one longer session (60–90 minutes) for mixed practice.

You can stretch this to 6 weeks simply by spreading topics out more.

Week 1 – Rebuild the Foundations (Arithmetic & Number Sense)

Goal: Get solid with basic arithmetic, fractions, decimals, and percents.

- **Day 1:**
 - Integers, order of operations.
 - 10 practice problems mixing positive/negative numbers and PEMDAS.
- **Day 2:**
 - Fractions (simplifying, add/subtract, multiply/divide).
 - 10–15 fraction problems.
- **Day 3:**
 - Decimals & percents (conversions, percent of a number).
 - 10 percent problems (sales, discounts, tax).

- **Day 4:**
 - Ratios, rates, proportions, basic distance = rate $\times$ time problems.
 - 8–10 word problems.
 - **Long session (Week 1 end):**
 - Timed mini-test: 10 AR-style + 10 MK-style problems using topics from week.
 - Review every mistake and write “what went wrong” in 1 sentence.
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Week 2 – Algebra Focus

Goal: Get comfortable with solving equations and manipulating expressions.

- **Day 1:**
 - Simplifying expressions, distribution, combining like terms.
 - 10–15 practice problems.
 - **Day 2:**
 - Solving linear equations (one-step, two-step, and multi-step).
 - 10 equations.
 - **Day 3:**
 - Inequalities; solving and graphing on a number line (rough sketch).
 - 10 inequality problems.
 - **Day 4:**
 - Systems of equations (substitution and elimination).
 - 6–8 problems.
 - **Long session (end of Week 2):**
 - Mix: 8 AR-style word problems that require algebra + 8 MK-style algebra problems.
 - Time yourself like on the test.
 - Review and categorize errors: content vs rushing vs misreading.
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Week 3 – Geometry + Mixed Word Problems

Goal: Lock in formulas and apply them in AR-style settings.

- **Day 1:**
 - Lines, angles, triangle basics.
 - Practice: 10 angle/triangle problems.
- **Day 2:**
 - Perimeter and area (rectangles, triangles, circles).
 - 10 geometry problems (some pure, some word problems).
- **Day 3:**
 - Volume (boxes, cylinders), Pythagorean Theorem.
 - 10 problems.

- **Day 4:**
 - Mixed word problems: distance, work, mixture, age.
 - 8–10 problems.
 - **Long session (end of Week 3):**
 - Timed set: 15 AR-style + 15 MK-style questions.
 - Try to keep pace ~1 question per minute or slightly faster.
 - Review: create a “formula sheet” from the problems—only the formulas that actually came up.
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Week 4 – Test Simulation & Polishing

Goal: Simulate AFOQT Quant pace and fix remaining weak spots.

- **Day 1:**
 - Take a **timed AR section** (25 questions in ~29 minutes from a practice source).
 - Score it; list top 3 weak topics.
- **Day 2:**
 - Take a **timed MK section** (25 questions in ~22 minutes).
 - Score it; list top 3 weak topics.
- **Day 3:**
 - “Error-only” session:
 - Go back over Weeks 1–3 and redo every question you missed without looking at the old work.
 - Make sure you can do each one correctly now.
- **Day 4:**
 - Short mixed drill: 5 AR + 5 MK under timed conditions (about 12–15 minutes).
 - Focus on staying calm and methodical even when the clock is running.
- **Long session (end of Week 4):**
 - Full Quant simulation:
 - AR section + short break + MK section, back-to-back.
 - Review only after everything is done.
 - Note: Did you panic? Run out of time? Rush and make silly mistakes?
 - Plan how you’ll handle that on the real test (e.g., “skip hard ones quickly,” “breathe every 5 questions,” etc.)